

Introduction to Commodities & Commodity Derivatives

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Part I

Commodity Fundamentals

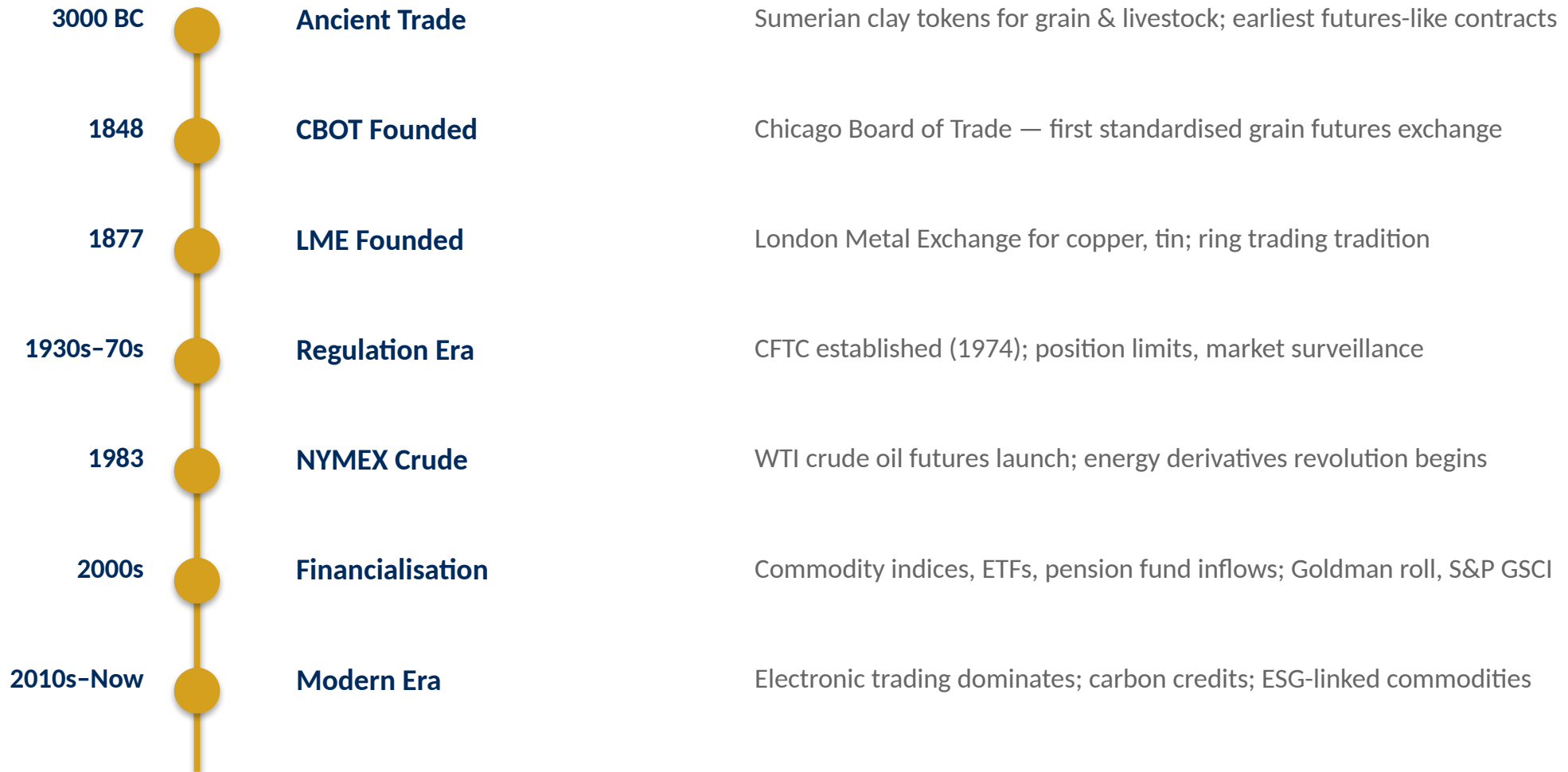
1. What Are Commodities?

- ▶ A commodity is a basic, standardised good used in commerce that is interchangeable with other goods of the same type
- ▶ Commodities are fungible — one unit equals another of the same grade and specification
- ▶ They form the building blocks of the global economy — energy, food, metals, raw materials
- ▶ Commodity prices are primarily supply/demand-driven, unlike equities which are earnings-driven
- ▶ Physical deliverability distinguishes commodities from purely financial assets
- ▶ Storage, transport, and quality grading create unique cost dimensions

Key Characteristics

Fungibility
Physical deliverability
Standardised grading
Supply-driven pricing
Storage costs matter
Strong seasonal patterns
Geopolitical sensitivity
Weather dependency (ags/energy)

2. History & Evolution of Commodity Markets



3. Commodity Asset Classes — Overview

Energy

Crude Oil (WTI, Brent)
Natural Gas, LNG
Power, Coal
Carbon Credits

Base Metals

Copper, Aluminium
Zinc, Nickel
Lead, Tin
(LME, COMEX, SHFE)

Precious Metals

Gold, Silver
Platinum, Palladium
Rhodium
(LBMA, COMEX)

Agriculture

Grains: Wheat, Corn
Softs: Coffee, Sugar
Livestock
(CBOT, ICE)

Weather & Environmental

HDD/CDD Futures
Catastrophe Bonds
Rainfall Indices
Carbon (EU ETS)

Global commodity futures turnover exceeds ~\$6 trillion annually. Energy accounts for ~40%% of total trading volume.

3a. Energy Commodities — Deep Dive

Crude Oil & Refined Products

- ▶ Benchmarks: WTI (NYMEX), Brent (ICE), Dubai (DME)
- ▶ Crack spreads: refining margin = product – crude
- ▶ Calendar spreads reflect storage / contango dynamics
- ▶ Geopolitical risk premium is significant
- ▶ OPEC+ supply decisions are a primary volatility driver

Natural Gas, Power & Carbon

- ▶ Henry Hub (US), TTF (Europe), JKM (Asia LNG)
- ▶ Extreme seasonality — heating/cooling demand cycles
- ▶ Power markets: locational marginal pricing (LMP)
- ▶ Carbon: EU ETS, CORSIA — growing regulatory asset class
- ▶ Weather derivatives hedge volumetric risk for utilities

3b. Metals — Base & Precious

Base Metals (Industrial)

- ▶ Traded on LME, COMEX, SHFE (Shanghai)
- ▶ Copper — 'Dr. Copper' as economic bellwether
- ▶ Aluminium — largest by volume on LME
- ▶ Unique LME structure: prompt dates, not monthly expiry
- ▶ Warehouse stocks & queues drive physical premiums
- ▶ Key driver: China industrial demand (~50%% global)

Precious Metals (Bullion)

- ▶ Gold — safe haven, central bank reserves, jewellery
- ▶ LBMA Gold Price: twice-daily electronic auction
- ▶ Silver — industrial + investment dual demand
- ▶ PGMs (Platinum, Palladium) — autocatalyst demand
- ▶ Gold leasing & forward rates (GOFO / swap rates)
- ▶ ETF flows are a major price driver

3c. Agriculture & Weather Derivatives

Agricultural Commodities

- ▶ Grains: Wheat, Corn, Soybeans (CBOT)
- ▶ Softs: Coffee, Sugar, Cocoa (ICE, NYBOT)
- ▶ Livestock: Live Cattle, Lean Hogs (CME)
- ▶ Strong seasonal planting/harvest cycles
- ▶ USDA reports are major event risk catalysts
- ▶ Weather, trade policy, biofuel mandates drive prices

Weather Derivatives

- ▶ HDD (Heating Degree Days) / CDD (Cooling Degree Days)
- ▶ CME-listed weather futures & options
- ▶ OTC structures for bespoke risk transfer
- ▶ Users: energy companies, agriculture, insurance
- ▶ Catastrophe bonds — insurance-linked securities
- ▶ Growing market driven by climate volatility

4. Physical vs Financial Commodity Markets

Physical Markets

- ▶ Spot / cash transactions for immediate delivery
- ▶ Governed by logistics, storage, quality grading
- ▶ Participants: producers, refiners, consumers
- ▶ OTC bilateral contracts dominate
- ▶ Price discovery via benchmarks (Platts, Argus, LBMA)
- ▶ Settlement in physical goods

Financial Markets

- ▶ Exchange-traded futures, options, swaps
- ▶ No physical delivery in most cases (cash-settled)
- ▶ Participants: banks, hedge funds, speculators
- ▶ Standardised contracts on CME, ICE, LME, TOCOM
- ▶ Mark-to-market daily with margin requirements
- ▶ Used for hedging, speculation & portfolio diversification

5. Key Market Participants

Producers & Extractors

Oil majors, mining cos,
farmers, utilities

Consumers & End-Users

Airlines, manufacturers,
food processors

Trading Houses

Glencore, Vitol, Trafigura,
Cargill, Mercuria

Financial Institutions

Banks, hedge funds,
asset managers

Exchanges & Clearinghouses

CME, ICE, LME, SGX,
LCH, CME Clearing

The commodity ecosystem is unique in that physical logistics, storage, and transportation create a tight linkage between financial and physical markets. Speculators provide liquidity while hedgers transfer risk.

6. Major Exchanges & Benchmarks

Key Exchanges

- ▶ CME Group (NYMEX + COMEX + CBOT) — largest global platform
- ▶ ICE (Intercontinental Exchange) — Brent, softs, power
- ▶ LME (London Metal Exchange) — base metals, unique date structure
- ▶ SHFE (Shanghai Futures Exchange) — China metals, rubber
- ▶ TOCOM (Tokyo) — precious metals, rubber, energy
- ▶ MCX (India) — gold, silver, crude, natural gas

Pricing Benchmarks

- ▶ WTI & Brent — global crude oil benchmarks
- ▶ Henry Hub — US natural gas
- ▶ LBMA Gold Price — twice-daily gold fix
- ▶ Platts / Argus — physical market price assessments
- ▶ Baltic Dry Index — shipping / freight cost indicator
- ▶ S&P GSCI, BCOM — investable commodity indices

7. Commodity Indices

- ▶ S&P GSCI — production-weighted; heavily tilted toward energy (~60%%)
- ▶ Bloomberg Commodity Index (BCOM) — diversified; caps sector weights
- ▶ DBIQ Optimum Yield — Deutsche Bank; optimised roll strategy
- ▶ Rogers International Commodity Index (RICI)
- ▶ Thomson Reuters / CoreCommodity CRB Index
- ▶ Roll yield is a critical component of total return for index investors
- ▶ Indices use 'excess return' (futures only) vs 'total return' (+ T-bill collateral)

Index Construction

Production weighting (GSCI)

Liquidity constraints

Sector caps for diversification

Roll schedule (front month vs optimised)

Excess vs Total Return

Rebalancing frequency

Part II

Pricing Fundamentals

8. Spot, Futures & the Forward Curve

Spot & Futures Basics

- ▶ Spot price: current market price for immediate delivery
- ▶ Futures price: agreed price for delivery at a future date
- ▶ Forward curve: term structure of futures prices across expiries
- ▶ Basis = Spot – Futures (or vice versa, convention varies)
- ▶ Calendar spread = price difference between two futures months

Pricing Relationships

- ▶ $F = S \times e^{\{(r + u - y)T\}}$
- ▶ r = risk-free rate (financing cost)
- ▶ u = storage cost
- ▶ y = convenience yield
- ▶ The forward curve shape encodes market expectations
- ▶ Seasonality is especially visible in nat gas & agriculture

9. Cost of Carry & Convenience Yield

- ▶ Cost of carry = financing cost + storage cost + insurance – convenience yield
- ▶ Convenience yield = benefit of holding physical inventory (optionality of immediate supply)
- ▶ When inventories are LOW, convenience yield is HIGH → backwardation
- ▶ When inventories are HIGH, convenience yield is LOW → contango
- ▶ The relationship determines the slope of the forward curve

Full Cost-of-Carry Formula

$$F(T) = S \times \exp[(r + u - y) \times T]$$

Where: S = spot price, r = risk-free rate, u = storage rate, y = convenience yield, T = time

If carry cost > convenience yield → Contango (upward sloping curve)

If carry cost < convenience yield → Backwardation (downward sloping curve)

10. Contango vs Backwardation

Contango

- ▶ Futures price > Spot price
- ▶ Forward curve slopes UPWARD
- ▶ Normal condition when storage is abundant
- ▶ Storage costs + financing dominate convenience yield
- ▶ Rolling futures generates NEGATIVE roll yield
- ▶ Example: Crude oil during inventory gluts
- ▶ WTI went negative in April 2020 due to extreme contango

Backwardation

- ▶ Futures price < Spot price
- ▶ Forward curve slopes DOWNWARD
- ▶ Indicates tight physical supply / high demand
- ▶ Convenience yield dominates carry costs
- ▶ Rolling futures generates POSITIVE roll yield
- ▶ Example: Copper during supply disruptions
- ▶ Often seen in metals & ags during shortages

Part III

Commodity Derivatives

11. Introduction to Commodity Derivatives

- ▶ A commodity derivative is a financial contract whose value is derived from an underlying commodity
- ▶ Purpose: hedging price risk, speculation, portfolio diversification
- ▶ Types: Futures, Forwards, Options, Swaps, Structured Products
- ▶ Exchange-traded (standardised) vs OTC (bespoke)
- ▶ Commodity derivatives differ from equity/FX derivatives due to:
 - Physical delivery mechanics
 - Storage & transportation costs embedded in pricing
 - Seasonality effects on volatility term structure
 - Convenience yield (no direct equivalent in financial assets)

Market Size (Approx.)

Global commodity derivatives:

Notional: ~\$2.5 trillion

Exchange-traded: ~60%%

OTC: ~40%%

Energy: ~50%% of notional

Metals: ~25%%

Agriculture: ~20%%

Other: ~5%%

12. Commodity Futures & Forwards

Exchange-Traded Futures

- ▶ Standardised contract size, grade, delivery location
- ▶ Daily mark-to-market with margin (initial + variation)
- ▶ Clearinghouse eliminates counterparty risk
- ▶ Most contracts closed out before expiry (no physical delivery)
- ▶ Examples: CL (WTI crude, 1000 bbl), GC (gold, 100 oz)
- ▶ Typical tenors: monthly, out to 5+ years for energy

OTC Forwards

- ▶ Bespoke terms: any volume, grade, delivery point, tenor
- ▶ Bilateral credit risk (mitigated via CSA / collateral)
- ▶ No daily margin (typically settled at maturity)
- ▶ Used by corporates for specific hedging needs
- ▶ Price may differ from exchange futures (basis risk)
- ▶ Growing trend toward central clearing (Dodd-Frank / EMIR)

13a. Commodity Options — Vanilla

Calls & Puts

- ▶ Call: right to BUY the underlying at the strike price
- ▶ Put: right to SELL the underlying at the strike price
- ▶ Premium = intrinsic value + time value
- ▶ Exercise style: American (most physical) / European (financial)
- ▶ Listed on CME, ICE — options on futures, not on spot
- ▶ Typical underlier: front-month or specific futures contract

Key Differences from Equity Options

- ▶ Underlier is a futures contract, not a stock price
- ▶ Black-76 model (not Black-Scholes) — no cost-of-carry in option formula
- ▶ Volatility smile is often asymmetric (skewed for energy)
- ▶ Strong seasonality in vol term structure
- ▶ Physical delivery possible on exercise
- ▶ Time spreads reflect calendar structure, not dividend timing

13b. Commodity Options — Exotics

- ▶ Asian Options — payoff based on average price over a period (very common in energy/metals)
- ▶ Barrier Options — knock-in / knock-out; cheaper premium; used in structured hedging
- ▶ Spread Options — payoff on price differential (crack spreads, spark spreads, location spreads)
- ▶ Swaptions — option to enter into a commodity swap at a fixed price
- ▶ Calendar Spread Options — option on the spread between two futures contract months
- ▶ Quanto Options — commodity payoff in a different currency (remove FX risk)
- ▶ Lookback Options — payoff based on max/min price during the option's life
- ▶ Accumulators / Decumulators — structured products with periodic fixing

Why Asians Dominate

Physical consumers/producers
hedge average purchase/sale price
Reduces volatility of hedge vs spot
Lower premium than vanilla
Averaging period matches
physical delivery schedule
Standard in energy OTC markets

14. Black-76 Pricing Model

The standard model for pricing European options on commodity futures

$$\begin{aligned}\text{Call} &= e^{-rT} [F \cdot N(d_1) - K \cdot N(d_2)] \\ \text{Put} &= e^{-rT} [K \cdot N(-d_2) - F \cdot N(-d_1)] \\ d_1 &= [\ln(F/K) + \frac{1}{2}\sigma^2 T] / (\sigma\sqrt{T}) \quad d_2 = d_1 - \sigma\sqrt{T}\end{aligned}$$

- ▶ F = futures price (replaces spot in Black-Scholes)
- ▶ K = strike price, T = time to expiry, σ = implied volatility
- ▶ No cost-of-carry term — futures price already incorporates it
- ▶ Assumes log-normal distribution of futures prices
- ▶ Key limitation: does not capture volatility smile/skew
- ▶ Extensions: SABR model, local vol, stochastic vol for smile fitting

Black-76 vs Black-Scholes

$$\text{BS: } C = S \cdot N(d_1) - K e^{-rT} \cdot N(d_2)$$

$$\text{B76: } C = e^{-rT} [F \cdot N(d_1) - K \cdot N(d_2)]$$

Key diff: F replaces $S e^{rT}$

Result: no carry term needed

15a. Commodity Swaps

Fixed-for-Floating Swaps

- ▶ Most common commodity derivative in OTC markets
- ▶ One party pays FIXED price, receives FLOATING (market) price
- ▶ Floating leg references a benchmark (Platts, NYMEX settle, etc.)
- ▶ Tenor: monthly, quarterly, or calendar year averaging periods
- ▶ Cash-settled — no physical delivery
- ▶ Used by producers to lock in revenue, consumers to lock in cost

Other Swap Types

- ▶ Basis Swaps — exchange one floating index for another (e.g. WTI vs Brent)
- ▶ Crack Spread Swaps — refiner margin hedge (crude vs products)
- ▶ Spark Spread Swaps — power plant margin (gas vs power)
- ▶ Index Swaps — total return on a commodity index
- ▶ Differential Swaps — location / quality basis hedging
- ▶ Increasingly centrally cleared (LCH, CME OTC)

15b. Commodity Structured Products

- ▶ Collars — buy a put + sell a call (or vice versa); zero-cost hedging strategy
- ▶ Three-Way Collars — add a sold put below the bought put to reduce cost further
- ▶ Extendible Swaps — swap with an embedded option to extend the tenor
- ▶ Accumulators — daily/periodic fixing; accumulate at discount; with knock-out barrier
- ▶ Target Redemption Forwards (TARFs) — accumulate until target profit reached
- ▶ Participating Forwards — forward with upside participation (embedded option)
- ▶ Commodity-Linked Notes — principal protection + commodity upside exposure
- ▶ Min/Max structures, Ratio spreads, Butterflies, Seagulls

Popular Hedging Combos

Producer (protect downside):

- Buy put (floor)
- Collar (floor + cap)
- Three-way collar

Consumer (protect upside):

- Buy call (cap)
- Reverse collar
- Participating forward

16. The Greeks in Commodities

Greek	Definition	Commodity-Specific Notes
Delta (Δ)	Sensitivity to futures price	Typically quoted as % of notional; varies by underlier
Gamma (Γ)	Rate of change of delta	Highest ATM near expiry; critical for short-dated energy
Vega (v)	Sensitivity to implied vol	Commodity vols are highly seasonal (nat gas winter > summer)
Theta (Θ)	Time decay per day	Accelerates near expiry; impacts short option books
Rho (ρ)	Sensitivity to interest rates	Smaller impact in B76 (discounting only, no carry in formula)

16b. Volatility in Commodity Markets

Volatility Characteristics

- ▶ Commodity vol is highly seasonal (e.g. nat gas: winter >> summer)
- ▶ Skew: energy typically has positive call skew (fear of spike)
- ▶ Mean-reversion in prices → vol term structure often inverted
- ▶ Event risk: OPEC meetings, USDA reports, weather events
- ▶ Realised vol spikes during supply disruptions
- ▶ Asian option vol < vanilla vol due to averaging effect

Vol Modelling Approaches

- ▶ Black-76 flat vol — simplest, single implied vol per strike/expiry
- ▶ SABR model — stochastic vol, captures smile dynamics
- ▶ Local volatility (Dupire) — deterministic vol surface
- ▶ Stochastic local vol (SLV) — hybrid approach
- ▶ Seasonal vol overlays for energy/agriculture
- ▶ Commodity vol surfaces: strike vs moneyness conventions vary

Part IV

Risk & Regulation

17. Risk Management & Hedging Strategies

- ▶ Hedging objective: reduce P&L volatility from commodity price movements
- ▶ Short hedge (producer): sell futures/buy puts to lock in sale price
- ▶ Long hedge (consumer): buy futures/buy calls to lock in purchase price
- ▶ Basis risk: mismatch between hedge instrument and actual exposure
- ▶ Rolling risk: cost of maintaining hedge across futures expiry cycles
- ▶ Cross-commodity hedging: e.g. jet fuel hedged with crude (crack spread risk)
- ▶ Hedge effectiveness testing under IFRS 9 / ASC 815 (accounting standards)

Common Risk Metrics

VaR (Value at Risk)
CVaR / Expected Shortfall
Greeks (Delta, Gamma, Vega)
Stress testing scenarios
Concentration risk
Liquidity risk (bid-ask, depth)
Delivery / operational risk
Basis point value (BPV)

18. Regulatory Landscape

US Regulation

- ▶ CFTC (Commodity Futures Trading Commission) — primary US regulator
- ▶ Dodd-Frank Act (2010): swap reporting, clearing mandates, position limits
- ▶ Position limits on speculative trading in 25 core commodities
- ▶ Swap Execution Facilities (SEFs) for price transparency
- ▶ Large Trader Reporting System (LTRS)
- ▶ Volcker Rule impacts on proprietary commodity trading

Global Regulation

- ▶ EU: MiFID II / MiFIR — position limits, commodity ancillary exemption
- ▶ EMIR — clearing obligation for OTC commodity derivatives
- ▶ UK: FCA post-Brexit commodity derivatives regime
- ▶ Basel III / FRTB — capital charges for commodity risk
- ▶ Environmental: EU ETS, Carbon Border Adjustment Mechanism (CBAM)
- ▶ ESG disclosure requirements for commodity portfolios

20. Key Takeaways

1

Commodities are physical assets

Unlike equities/FX, physical delivery, storage, and transport fundamentally shape pricing

2

Forward curve is king

Contango / backwardation / seasonality drive trading strategies and roll returns

3

Black-76 is the workhorse

Standard model for commodity options — futures-based, no cost-of-carry in formula

4

Derivatives serve real hedging needs

Producers and consumers use swaps, options, and structures to manage price risk

5

Volatility is seasonal

Energy and agriculture vol term structures reflect weather and harvest cycles

6

Regulation is evolving

Dodd-Frank, MiFID II, FRTB, and carbon markets are reshaping the landscape

Recommended Resources

Books & References

- ▶ "Commodity Derivatives" — Neil Schofield (Wiley)
- ▶ "Commodities and Commodity Derivatives" — Hélyette Geman
- ▶ "The Handbook of Commodity Investing" — Fabozzi et al.
- ▶ "Trading and Exchanges" — Larry Harris
- ▶ "Options, Futures, and Other Derivatives" — John C. Hull
- ▶ CME Group Education Centre (cmegroup.com/education)

Industry Resources

- ▶ CFTC.gov — Commitments of Traders (COT) reports
- ▶ ICE / LME / CME websites — contract specifications
- ▶ Platts / Argus — physical price assessments
- ▶ EIA (US Energy Information Administration) data
- ▶ World Gold Council — gold demand/supply statistics

Questions & Discussion

Thank you!

Vaibhav Gupta